

The Sentinel System: An Assistant for the Overloaded Clinician

A 500-word overview for a sports psychologist

Word count: 511

1. The Problem - Buried, Not Absent

Mental health clinicians are not failing their patients; they are drowning. Patients spend 167 of every 168 hours out of sight - risk builds in those unseen hours and surfaces only at the next appointment. The clinician has no time to look: documentation, journal review, and manual risk assessment consume hours each week, and India runs 0.75 psychiatrists per 100,000 against a WHO recommendation of 1 per 10,000. Telehealth is appointment-only; wellness apps are generic. Between-session hours stay invisible, so risk is discovered late and crises land without warning.

2. Two Signals, One Picture

Sentinel compares what a patient says with what their body reports, and flags when they disagree.

The subjective channel - words. Every journal entry is read, summarized, and classified across 28 emotion labels by a classifier trained on 54,263 real GoEmotions examples. Keywords are negation-aware, so "I'm not okay" is never read as positive.

The objective channel - the body. A companion ring streams biomarkers over a securely paired (token-fingerprinted) connection: heart rate, heart-rate variability (HRV, with RMSSD/SDNN), stress level, sleep hours, and blood-oxygen saturation.

The discrepancy engine. When the channels disagree, Sentinel notices. A journal that reads positive while HRV is low and resting heart rate is high is a red flag - the patient may be masking, and the risk assessment must include that tension, not just the words. This comparison runs in ~0.1 ms and broadcasts an alert the moment a mismatch is detected, so the clinician sees the contradiction at a glance instead of discovering it weeks later in a chart.

3. AI and Hardware Working Together

The stacked signals feed the risk engine, which reports a numeric score with a plain-language explanation of which signals contributed - never a black box. When a crisis triggers, a staged protocol acts automatically: an immediate siren with an explainable reason, a trusted-contact email with a signed, expiring link at 30 seconds, and helpline escalation at 60 seconds. The clinician halts it with one tap and the timer freezes; they are never the single point of failure. Self-monitoring metrics also surface the clinician's own stress and fatigue, guarding against secondary trauma in the person holding every patient's story.

4. Judgment Stays Human

The AI summarizes, triages, and alerts - it never diagnoses or prescribes. Every output carries its model and confidence, and nothing enters a clinical record without clinician approval. The AI runs locally, so patient data never leaves the clinic.

5. Validation and Cost

Sentinel passes 99/99 automated tests, detects discrepancies with 96% accuracy in ~0.1 ms, and ships a 47-run timing benchmark - all reproducible. It runs on free infrastructure at a \$0 monthly cost, deployable in any clinic, school, or community center.

6. The Promise

Sentinel does not replace clinical judgment. It gives it back the time it lost - hours each week, a safety net that never sleeps, and the ability to see the patient whose words and body disagree before the words turn into a crisis.